

Sample Problem — Control Resolution ($CR = p / n$)

STAR MILL axis: lead-screw pitch **p = 4 mm**, stepper motor **n = 200 steps/rev**.

$$CR = p / n = 4 \text{ mm} / 200 = 0.02 \text{ mm} = 20 \text{ micrometre}$$

The smallest increment of motion the axis can command is 20 μm per step.

Review Questions

1. What are the main limitations of traditional NC systems that led to the development of modern CNC machines?

Ans. Traditional NC used **hardwired control logic and punched paper/mylar tapes** to store the program. Its main drawbacks were: the program **could not be edited at the machine** — the whole tape had to be re-punched for even a small change; the **tapes wore out, tore and read unreliably**, causing scrap; there was **no program storage, memory or diagnostics**; the machine had to **re-read the tape for every cycle**, which slowed production; and because the control was hardwired, changing the machine's function needed physical rewiring. **CNC overcame all of these** by using an onboard microcomputer (MCU) that stores the program in memory and is "soft-wired", so programs can be edited at the machine, stored in a library, run with on-board diagnostics and interpolation, and executed with far higher reliability, accuracy and productivity.

2. Explain the critical difference between machine accuracy and machine precision using a target diagram analogy.

Ans. **Accuracy** is how close the machine gets to the **actual target (commanded) point** — i.e. hitting the bull's-eye. **Precision (repeatability)** is how closely **repeated moves cluster together**, whether or not that cluster is on the target. Using a target: a tight group placed away from the centre is **precise but not accurate**; shots scattered all around the centre are **accurate on average but not precise**; and a tight group sitting on the centre is **both accurate and precise**. In practice, accuracy includes **systematic errors** (which can be measured and calibrated out), whereas precision reflects **random scatter** (which cannot be calibrated away). A good CNC machine must be both accurate and precise.

3. How does Intelligent Manufacturing adapt to production variations compared to standard flexible automation?

Ans. Standard **flexible automation** can make a variety of products, but it must be **reprogrammed or reset by a human** whenever the product or mix changes. **Intelligent Manufacturing** goes a step further: using sensors, real-time data, IoT connectivity, AI/machine-learning and adaptive closed-loop control, it **detects variations by itself** — tool wear, material differences, temperature drift or dimensional error — and **automatically adjusts the process parameters** to correct them, often supported by a digital twin and predictive maintenance. It is therefore **self-monitoring and self-optimising**, adapting to production variation in real time instead of waiting for a human to change the setup.

4. What role does a Direct Numerical Control (DNC) network play in an automated production facility?

Ans. In **DNC**, a central computer is networked to a group of CNC/NC machines and acts as the hub for part-program data. It **stores all the part programs centrally, distributes the required program to each machine on demand** (removing individual tape readers and manual loading), and **collects production and machine-status data back** from the shop floor for monitoring. It also manages **program version control and back-up**, and links the machines upward to scheduling, MES and ERP systems. In this way DNC ties isolated machines together into a coordinated, data-driven **computer-integrated manufacturing (CIM)** environment.

Sample Problem — Linear Displacement ($x = N/E \times P$)

Rotary encoder **E = 2500 pulses/rev**, ball-screw pitch **P = 5 mm/rev**, pulses counted **N = 12 500**.

$$x = (N / E) \times P = (12500 / 2500) \times 5 = 5 \times 5 = 25 \text{ mm}$$

The table moves a linear distance of 25 mm.

Assignment Questions

1. How do recirculating ball screws eliminate the problem of "backlash" found in traditional Acme lead screws?

Ans. In a ball screw, hardened steel **balls recirculate between the screw shaft and the nut**, so motion is transmitted through **rolling contact** instead of the sliding contact of an Acme thread. Backlash (the lost motion felt when an axis reverses) is removed by **preloading** — using a **double nut, an oversized ball set, or a lead shift** so that the balls are held in firm contact with both flanks of the thread at all times, eliminating the internal clearance. This gives **near-zero backlash**, much higher mechanical efficiency (about 90 % against roughly 40 % for an Acme screw), higher rigidity and far better positioning accuracy and repeatability.

2. Compare the functional advantages of Linear Motion guideways versus Hardened and ground guideways in high-speed machining centres.

Ans. **LM (linear-motion, rolling) guideways** use recirculating balls or rollers, giving **very low friction, no stick-slip, very high rapid-traverse speeds, low heat generation and easy replacement** — ideal for fast, precise positioning. **Hardened and ground (box) guideways** are a large sliding-contact surface, giving **high load capacity, high stiffness and excellent vibration/chatter damping**, but with higher friction, more heat and lower speed. In **high-speed machining centres LM guideways are generally preferred** for their speed and precision, while box ways are retained on axes that take **heavy or interrupted roughing cuts**, where their damping protects the tool and surface finish.

3. Why might a high-precision CNC machine use Linear Glass Scales for positional feedback instead of relying solely on the servo motor's rotary encoder?

Ans. A rotary encoder on the motor or screw only measures the **screw's angular rotation**, so the table position is **inferred indirectly** from the assumed lead. It therefore **cannot detect any error that occurs after the screw** — ball-screw pitch error, thermal expansion of the screw, backlash, nut wear or coupling wind-up. A **linear glass scale** is mounted on the axis itself and measures the **table's true linear position directly** (a genuine closed loop that satisfies the Abbe principle), so it **sees and corrects all of those errors**. For micron-level accuracy — especially where the screw heats up during long runs — direct glass-scale feedback is essential.

4. What is the primary difference between Hydrostatic and Hydrodynamic bearings, and how does this affect spindle startup?

Ans. In a **hydrodynamic** bearing the supporting oil film is **generated by the rotation of the journal itself**; at rest and at low speed there is **metal-to-metal contact**, so there is friction and wear at every start and stop until the spinning shaft drags the film up.

In a **hydrostatic** bearing the oil is **supplied under pressure by an external pump**, so a **full supporting film exists even at zero speed**, giving a frictionless, **zero-wear start** and very high stiffness and accuracy. The trade-off is that the hydrostatic design needs a pump, filtration and more cost, whereas the hydrodynamic bearing is simpler and cheaper and performs well once it is running at steady high speed.

Mathematical Drill — Spindle Speed from Cutting Speed

Cutting velocity $v = (\pi \cdot D \cdot N) / 1000$, rearranged for spindle speed N :

$$N = (1000 \times v) / (\pi \times D)$$

v = cutting speed (m/min), D = work/cutter diameter (mm), N = spindle speed (RPM)

Assignment Questions

1. Explain why tool nose radius compensation (G41/G42) is vital when turning tapered profiles or circular contours on the XL TURN lathe.

Ans. A turning tool is programmed at its **theoretical sharp point**, but the real cutting edge is a **rounded nose of finite radius**. On straight turning and facing the contact point stays at that programmed point, so there is no error; but on **tapers and circular contours the actual contact point shifts around the curved nose**, so the part comes out **undercut or oversize** by an amount that depends on the slope and the nose radius. **G41/G42 nose-radius compensation** tells the control the nose radius and the tool orientation (the imaginary tool-tip quadrant), and it **automatically offsets the tool-centre path** so the programmed profile is reproduced exactly on the finished part. Without it, angled and curved features must be re-calculated by hand.

2. What occurs on a CNC mill if a programmer accidentally swaps a G02 command for a G03 path when milling a corner arc?

Ans. **G02 is clockwise** and **G03 is counter-clockwise** circular interpolation. If they are swapped, the control takes the same start point, end point and radius but **sweeps the arc in the opposite direction**, producing a **mirror-image / incorrect arc**. The cutter then either **gouges into the finished part** or cuts air on the wrong side of the profile, leaving material behind and ruining the geometry, and it can cause a **tool or fixture collision**. If the arc is defined by I/J (centre) or R and the wrong direction makes the geometry impossible, the control will throw a **radius/end-point alarm** and stop.

3. Differentiate between the physical coordinate referencing modes of G90 and G91 commands.

Ans. **G90 (absolute mode)** references **every coordinate from one fixed work origin (part zero)** — each value written is the actual position the tool must reach. **G91 (incremental mode)** references **each coordinate from the tool's current position** — each value is the distance and direction to move for that step. Absolute programming is easier to check against the drawing and recovers safely after an interruption, so it is used for most programs; incremental programming is convenient for **repeated or patterned features** (for example drilling an equally spaced row of holes) where the same relative move is repeated.

4. What primary parameter determines whether a feed rate command uses feed-per-minute (G94) or feed-per-revolution (G95) parsing modes?

Ans. It is the **reference unit chosen for the feed rate** — whether the feed is tied to **time** or to **spindle rotation**. **G94 = feed per minute (mm/min)**, where the axis advances a set distance each minute regardless of spindle speed; this is normal for **milling**. **G95 = feed per revolution (mm/rev)**, where the axis advances a set distance for every revolution of the spindle; this is normal for **turning**, because tying the feed to

each revolution keeps the **chip load and surface finish consistent** as the cutting diameter (and hence spindle speed under constant surface speed) changes.

Problem Case — Spindle Speed from the Machinability Database

Database cutting speed **V_c = 150 m/min**, rough-turning a mild-steel rod of **D = 40 mm**:

$$\begin{aligned} N &= (1000 \times V_c) / (\pi \times D) = (1000 \times 150) / (\pi \times 40) \\ &= 150000 / 125.66 = \text{approx. } 1194 \text{ RPM} \end{aligned}$$

Questions

1. Why is CAPP considered an indispensable link when transitioning a factory floor from basic islands of automation to a fully computer-integrated manufacturing (CIM) system?

Ans. **CAPP (Computer-Aided Process Planning)** is the **bridge that connects CAD (design) to CAM (manufacturing)**. In a shop with only isolated "islands of automation" — standalone CAD, standalone CNC, standalone inspection — each system holds its own data and **cannot pass information automatically to the next**, so an engineer must manually translate the design into a process plan, which is slow and inconsistent. CAPP **automatically converts the design data into standardised, repeatable process plans and routing sheets** and feeds that consistent data forward to scheduling, MRP, the CNC machines and material handling. It therefore removes the manual gap, forms the **digital thread** through the factory, and is essential to knit the separate islands into a true **computer-integrated manufacturing (CIM)** system.

2. Under what specific manufacturing volume and variety scenarios would a company choose a Retrieval (Variant) CAPP system over a Generative CAPP system?

Ans. **Retrieval (Variant) CAPP** is chosen for **high production volume with low part variety**, where most new parts are **similar to existing ones and fall into established part families** (coded, for example, with an Opitz system). The system simply **retrieves the family's master process plan and lightly edits it**, which is fast, cheap and reliable, and it works well in a stable product line. **Generative CAPP**, by contrast, suits **high variety / low volume** or genuinely **new, unique parts** for which no master plan exists, because it **builds the plan from scratch** using logical rules and a machinability database — more flexible but far more complex and costly to develop. So a company making **many standardised, repetitive parts** picks the Variant system.

Analytical Drill — AGV Cycle Time & AS/RS Capacity

Part A — AGV cycle time: $v_c = 60$ m/min, $L_d = 120$ m, $L_e = 90$ m, $T_l = 0.8$ min, $T_u = 0.7$ min.

$$\begin{aligned} T_c &= T_l + T_u + L_d/v_c + L_e/v_c \\ &= 0.8 + 0.7 + 120/60 + 90/60 \\ &= 0.8 + 0.7 + 2.0 + 1.5 = 5.0 \text{ minutes} \end{aligned}$$

Part B — AS/RS storage capacity: Total Storage Capacity = $2 \cdot n_a \cdot x \cdot y$ (n_a = number of aisles, x = horizontal compartments, y = vertical tiers; the factor 2 accounts for a rack on **both sides** of every aisle).

Assignment Questions

1. Which Principle of Material Handling focuses specifically on reducing dead-weight and maximising space utilisation in a facility?

Ans. The **Space Utilisation Principle**. It states that all available space in a facility must be used as effectively as possible — making the best use of the building's **cubic volume** rather than just its floor area, chiefly by storing **vertically in high-density racks** (as an AS/RS does) and by minimising wasted aisles and dead volume. Maximising cube utilisation is its core aim; the related **Unit-Load Principle** complements it by keeping the **dead-weight of each handling unit low** relative to the payload carried, so that less energy and equipment are spent moving the container rather than the product.

2. Explain how a vehicle collision-avoidance loop works when managing a fleet of multiple AGVs on a shared track layout.

Ans. The shared guide path is divided into **zones or blocks**, and a **central traffic controller (zone/blocking control)** allows **only one AGV inside a block at a time**. Before an AGV may enter the next block it must **request permission**; the controller grants it only if that block is **free**, otherwise the vehicle **stops and waits** until the block clears. This blocking logic is backed up by **onboard proximity, laser and bumper sensors** and by fixed **right-of-way rules at intersections and merges**. Together they guarantee that two vehicles can never occupy the same path segment, which **prevents collisions and also avoids deadlock** (where vehicles mutually block each other) on a busy shared layout.

3. What are the primary performance advantages of implementing an AS/RS compared to a conventional manual forklift-and-pallet rack system?

Ans. An **AS/RS** gives **much higher storage density**, because its tall, narrow aisles and vertical racking use the building's cubic space far better than the wide aisles a forklift needs. It provides **faster, automated, 24/7 storage and retrieval** with higher and more consistent throughput; **greater accuracy and real-time inventory control** through its warehouse-management software, so items are rarely lost or mis-picked; **reduced labour and labour cost; improved safety**, since there are far fewer human-forklift accidents; and better, software-enforced **stock rotation (FIFO)**. A manual forklift-and-pallet system, by comparison, is slower, space-hungry, labour-intensive and more error-prone.

4. Why are 2D matrix codes (such as QR codes) increasingly favoured over legacy 1D linear barcodes in harsh automated manufacturing environments?

Ans. 2D matrix codes store far more data in a much smaller area than a 1D barcode, and they can be **read from any orientation (omnidirectionally)** rather than needing a straight-line scan. Most importantly, they contain **built-in error-correction (redundancy)**, so a code can still be **decoded even when it is partly damaged, dirty, scratched, faded or poorly printed**. In harsh manufacturing conditions — dust, oil, coolant, heat and abrasion — a 1D barcode fails as soon as a few of its bars are smudged or worn, whereas a 2D code **remains reliably readable**, which makes it far more robust for automated tracking.

Sample Problem — Mean Positioning Error of a CMM

A 50.000 mm gauge block is measured four times: 50.002, 50.005, 50.003, 50.002 mm.

$$\begin{aligned}\text{Mean } \mu &= (50.002 + 50.005 + 50.003 + 50.002) / 4 \\ &= 200.012 / 4 = 50.003 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Mean positioning error } E_{\text{pos}} &= | \mu - X_{\text{nom}} | \\ &= | 50.003 - 50.000 | = 0.003 \text{ mm} = 3 \text{ micrometre}\end{aligned}$$

$$\text{Repeatability } \sigma = \sqrt{\sum (x_i - \mu)^2 / (n - 1)} = \text{approx. } 1.41 \text{ micrometre}$$

Technical Questions

1. Compare contact CMM inspection methods with non-contact laser scanning in terms of measuring speed, point-cloud density, and accuracy limits.

Ans. A **contact (touch-trigger) CMM** physically touches the surface and records **discrete points one at a time**, so it is comparatively **slow and gives a low point density**, but each point is captured to **very high accuracy** (micron level) and is unaffected by surface colour or finish — making it best for **precise prismatic features** such as bores, hole positions and distances. **Non-contact laser scanning** sweeps a laser line or point cloud over the part and captures **thousands of points per second**, giving **extremely high speed and very dense point clouds** that are ideal for **complex free-form 3D surfaces and reverse engineering**; however its **accuracy is lower** (typically tens of microns) and it is **sensitive to surface finish, colour and reflectivity** (shiny or dark surfaces scatter the beam). In short: contact CMM = slow, sparse, highly accurate; laser scanning = fast, dense, less accurate.

2. Explain how environmental conditions (such as ambient room-temperature variations) introduce measurement errors in high-precision coordinate metrology.

Ans. Metals **expand and contract with temperature** (thermal expansion). Dimensional metrology is internationally referenced to **20 °C**, so whenever the room differs from 20 °C, both the **workpiece** and the **machine's scales and structure** change size, and the measured value **drifts away from the true dimension**. **Temperature gradients** — a warm side and a cool side, or heat from motors and lighting — are worse still, because they **distort the CMM frame unevenly** and bend the axes. Since even a **1 °C change produces micron-level errors** over typical part sizes, high-precision metrology is carried out in **temperature-controlled rooms held at 20 ± 1 °C**, parts are given time to **soak to room temperature** before measurement, and software **temperature compensation** is applied to remove the residual error.